Weekly Assignment: Graph Theory
(Epp 10.1, 10.2, & 10.3)

Complete each of the following problems. Show all work and justify your answers.

Problem 1 (Epp 10.1 #2). In the graph below, determine whether the following walks are trails, paths, closed walks, circuits, simple circuits, or just walks.

a. $v_1e_2v_2e_3v_3e_4v_4e_5v_2e_2v_1e_1v_0$

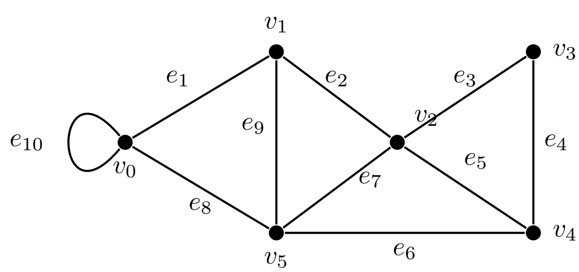
b. $v_2v_3v_4v_5v_2$

c. $v_4v_2v_3v_4v_5v_2v_4$

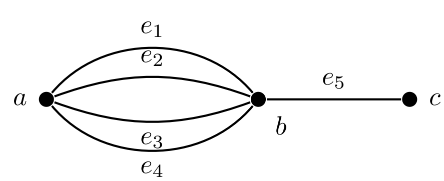
d. $v_2v_1v_5v_2v_3v_4v_2$

e. $v_0v_5v_2v_3v_4v_2v_1$

f. $v_5v_4v_2v_1$



Problem 2 (Epp 10.1 #5). Consider the following graph.

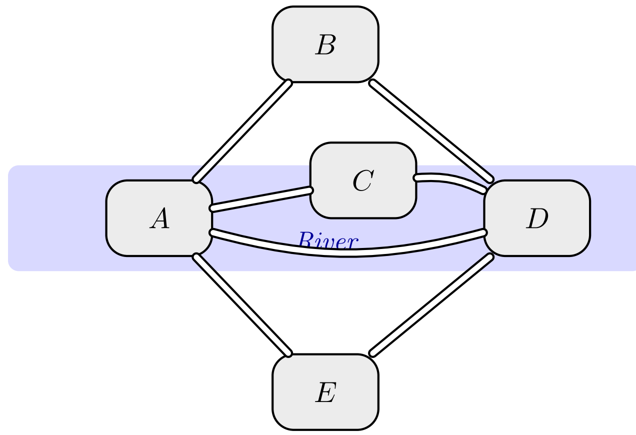


a. How many paths are there from a to c?

b. How many trails are there from a to c?

c. How many walks are there from a to c?

Problem 3 (Epp 10.1 #18). Is it possible to take a walk around the city whose map is shown below, starting and ending at the same point and crossing each bridge exactly once? If so, how can this be done?



Problem 4 (Epp 10.2 #5). Find graphs that have the following adjacency matrices.

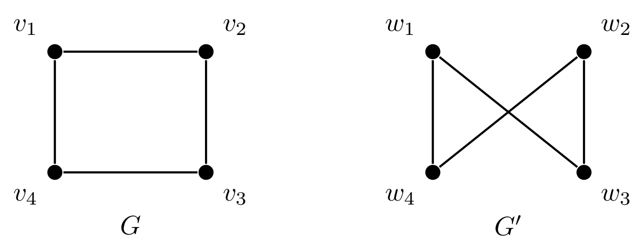
a. $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{pmatrix}$ **b.** $\begin{pmatrix} 0 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Problem 5 (Epp 10.2 #7). Suppose that for every positive integer i, all the entries in the ith row and ith column of the adjacency matrix of a graph are 0. What can you conclude about the graph?

Problem 6 (Epp 10.2 #9 b,c). Find each of the following matrix products.

b. $\begin{pmatrix} 2 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 5 & -4 \\ -2 & 2 \end{pmatrix}$ **c.** $\begin{pmatrix} -1 \\ 2 \end{pmatrix} \begin{pmatrix} 2 & 3 \end{pmatrix}$

Problem 7 (Epp 10.3 #7). Determine whether the simple graphs G and G' below are isomorphic. If they are, give a function $g\colon V(G) \to V(G')$ that defines the isomorphism. If they are not, give an invariant for graph isomorphism that they do not share.



Problem 8 (Puzzle). What is the maximum number of edges a simple disconnected graph with n vertices can have? Prove your answer.

Hint: If the graph is disconnected, it has at least two connected components. Which partition of the n vertices into components allows the most edges?